

COLLECTIVE DISCRIMINATION FROM CLASS-CORRELATED REPRESENTATION ERRORS IN INTERACTING NEURAL AGENTS

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ABSTRACT

Learners embedded in a population adapt not only what they predict but whom they learn from, so a representation error in one agent can restructure the information flow of the whole group. We study a population of neural agents whose perceptron opinions and directed trust variables co-adapt through an entropic approximation to Bayesian online learning derived for a noisy student–teacher channel. A fraction f_d of agents extend their representation of an incoming message with one coordinate that carries no information about the issue and depends only on the sender’s class, shifting the agreement field that drives their update by a strength d . We show that this shift translates the local opinion–trust flow rigidly, breaking an exact $\mathbb{Z}_2$ symmetry of the unbiased dynamics and displacing a separatrix we obtain in closed form, and we derive a parameter-free prediction $C_{CT} = f_d G(d)$ for the trust–class correlation. Replicated simulations exceed it by **[residual]**, which measures the amplification the population itself contributes. Sweeping (d, f_d) with **[24]** replicates per point reveals **[n]** collective regimes, separated by crossovers rather than transitions; magnitude-matched controls whose sign is uncorrelated with class establish that the effect requires class correlation and not merely added perturbation. Varying the agenda complexity $\alpha = P/K$ reverses whether affective or ideological structure forms first, crossing at $\alpha = \textbf{[crossover]}$. We further prove that the apparent transition at $d = 0$ is a forced zero-crossing of an odd response rather than a bifurcation: the affective sector has no linear restoring force at the class-symmetric state, so no critical field strength exists at any population size.

1 INTRODUCTION

Representations carry features that the task does not need. When one of those features is correlated with a group label, the usual consequence is a biased prediction, and the usual remedy is to audit or retrain the model that makes it. That framing assumes the learner is alone. Increasingly it is not: systems are assembled from many learned agents that read one another’s outputs, form some judgement of how far to rely on each other, and revise accordingly. In such a population a learner adapts not only *what it predicts* but *whom it learns from*, and a representation error acquires a second route of influence. It can reshape the information flow of the whole population.

The collective consequence of that second route is what we study. Online Bayesian and entropic accounts describe how a single learner should update from a noisy source (Oppen, 1996; 1998; Solla & Winther, 1999; Caticha, 2020); opinion-dynamics and signed-network models describe how populations polarise and how sign structures organise (Deffuant et al., 2000; Castellano et al., 2009; Heider, 1946; Cartwright & Harary, 1956); fairness research characterises task-irrelevant and group-correlated features in representations (Dwork et al., 2012; Hardt et al., 2016; Barocas et al., 2023). What is missing is the combination: what a group-correlated representation error does to a population whose *trust* co-adapts alongside its opinions.

The model, in plain language. A population of agents discusses a fixed agenda of issues. Each agent is a perceptron over a shared embedding of issues, so two agents can agree on some and disagree on others. Each agent additionally maintains, for every other agent, an estimate of how unreliable that agent is as a source — a trust, which is inferred rather than posited. Both sectors co-adapt through a single update derived from a Bayesian step on the agent’s belief. Every agent also carries an inert binary label. A fraction f_d of agents extend their representation of an incoming *message* with one extra coordinate that carries no information about the issue and depends only on the sender’s label, shifting their perceived agreement by $\pm d$ according to whether the sender shares it.

The mechanism, in one paragraph. That shift translates the local opinion–trust flow rigidly, and the translation moves a separatrix that we can locate exactly. One consonant basin — “agree and trust” — grows at the expense of the other, and it grows towards the in-group and shrinks towards the out-group. Crucially the shift enters the two sectors asymmetrically: it can flip the *sign* of a trust update, but only rescales the *magnitude* of an opinion update. Class structure therefore appears in trust first and reaches opinion only through it. That asymmetry predicts both the sign and the ordering of the effects we then measure.

Contributions.

- **A formal model of class-correlated representation error** (§4). We derive the field equation 8 from an explicit augmented representation of an interaction, state the assumptions under which the reduction is exact, and make precise the sense in which the added feature is irrelevant to the task.
- **An exact local analysis** (§4.1). The field translates the phase portrait rigidly; the separatrix is a line we give in closed form; the unbiased flow has a $\mathbb{Z}_2$ symmetry that the field breaks. No expansion in d is used anywhere.
- **A parameter-free prediction, and a measurement of what exceeds it** (§4.2). A basin-capture argument gives $C_{CT} = f_d G(d)$ with G closed-form at the model’s own initialisation. The measured correlation exceeds it by [residual], and that residual is the population’s own amplification.
- **A replicated map of the collective regimes** (§6), with regimes defined by a stated classification rule against null-calibrated thresholds, boundary uncertainty shown rather than asserted, and controls establishing that the effect requires class *correlation* and not merely a perturbation of the same size.
- **A temporal result** (§7): the agenda complexity $\alpha = P/K$ controls whether affective or ideological structure forms first, crossing over at $\alpha = [\text{crossover}]$.

A negative result we state plainly. The sign change of the trust–class correlation at $d = 0$ looks like a transition and is not one. We show that the affective sector has no linear restoring force at the class-symmetric state, so no critical field strength for C_{CT} exists at any N ; the sign change is a forced zero-crossing of an odd response. This explains why an earlier attempt to obtain a critical curve for this model returned a largest eigenvalue that was identically zero, and it redirects the search for a genuine threshold to the onset of *ideological* alignment, where we look for one.

Scope. This is a mechanistic model of interacting learners, not a model of people. Its agents are perceptrons, its groups are arbitrary inert labels, and its “tolerance” is a scalar. We draw no conclusions about human societies. What the results support is a claim about learning systems: a locally well-motivated update, deployed in a population, can convert an informationally empty label into a stable architecture of group distrust — a failure of the interaction rather than of the training data, and one invisible to any evaluation that inspects agents one at a time.

2 RELATED WORK

Online Bayesian and entropic learning. The update we use is the optimised online algorithm for the student–teacher problem, given a Bayesian reading by Oppen (1996; 1998) and Solla & Winther (1999) and an entropic-dynamics formulation by Caticha (2020); the identification of the inferred

channel-noise level with distrust, and its promotion to a dynamical variable, is due to Alves & Caticha (2016). We inherit the rule and claim nothing about it. What is new here is what it does to a population when one class-correlated coordinate is added to the representation, and the analysis of that is ours.

Neural-agent opinion and trust dynamics. Representing agents as classifiers over an issue space, rather than as scalars or spins, makes partial agreement possible (Caticha & Vicente, 2011; Caticha et al., 2015), which is what lets one ask whether distrust follows disagreement or the reverse. Earlier work in this family studies societies without any group label (Simões & Caticha, 2018; Caticha & Alves, 2019). The label, the representation error that carries it, and the collective consequences are the subject here.

Opinion dynamics and structural balance. Models of consensus and polarisation run from bounded-confidence rules (Deffuant et al., 2000; Hegselmann & Krause, 2002) through cultural dissemination (Axelrod, 1997) to reinforcement-driven echo chambers (Baumann et al., 2020); see Castellano et al. (2009) for a survey. Our triple-level diagnostics are those of structural balance (Heider, 1946; Cartwright & Harary, 1956; Antal et al., 2005; Marvel et al., 2011), but with one difference worth stating: there the sign structure is the primitive object, whereas here both sign structures are *generated* by inference. We also study a transient at a fixed horizon rather than an equilibrium, because the dynamics anneals and has no stationary state.

Fairness, sensitive attributes and spurious features. The dominant framing locates group bias in data and in single-model decisions, as representational harm (Blodgett et al., 2020; Bender et al., 2021) or as allocative unfairness with formal criteria attached (Dwork et al., 2012; Hardt et al., 2016; Barocas et al., 2023). Our mechanism is complementary: the outcome is produced by an irrelevant feature interacting with the *learning rule* across a population, and it is invisible to an audit that inspects agents one at a time. That irrelevant features degrade a classifier is old; that they can restructure a society of classifiers is the point here.

Collective effects of local rules. The general phenomenon — locally sensible optimisation producing undesirable system-level behaviour — is recognised across multi-agent learning, but we are not aware of a treatment in which the coupling runs through an *inferred* trust between learners. That is the gap this paper occupies.

3 A SOCIETY OF AGENTS THAT LEARN FROM EACH OTHER

Agents, issues, opinions. A population of N agents discusses an agenda of P issues. An issue is a unit vector $\hat{x} \in \mathbb{R}^K$: the embedding of an assertion in the K dimensions relevant to the question at hand. All agents embed issues identically and differ only in what they make of them, which separates the *representation* of an issue from the *opinion* an agent holds about it. Agent I carries a perceptron with weights $w_I \in \mathbb{R}^K$ and states $\sigma_I = \text{sign}(w_I \cdot \hat{x}) \in \{\pm 1\}$. This is the least structure that lets two agents agree on some issues and disagree on others; a model in which each agent is a single spin cannot, and graded disagreement turns out to drive the collective behaviour we report. Each agent also carries an immutable label $\kappa_I = \pm 1$, visible to others, splitting the population into two equal classes. *The label carries no information about any issue*; we make that precise in §4.

An agent’s state has two sectors. The *ideological* sector is a Gaussian belief over its own weights, mean $\hat{w}_I$ and covariance C_I . The *affective* sector is a Gaussian belief, held separately about every other agent J , over how unreliable J is as a source: mean $\mu_{J|I}$, the *distrust* I assigns to J , and variance $V_{J|I}$. Writing the joint belief as a product of Gaussians,

$$Q(w_I, z_I | \lambda_I) = \mathcal{N}(\hat{w}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \quad (1)$$

the dynamical variables are $\lambda_I = \{\hat{w}_I, C_I, \mu_{J|I}, V_{J|I}\}$. The means fix what an agent thinks; the variances fix how firmly, and thereby how fast it can be moved.

One interaction. At each step an issue $\hat{x}$, an emitter e and a receiver r are drawn uniformly. The emitter states σ_e ; the receiver treats it as a report of an unknown true label transmitted through a

channel whose flip probability is precisely what its distrust of the emitter encodes. Two scaled fields carry everything the update depends on:

$$h_w = \frac{\hat{w}_r \cdot \hat{x} \sigma_e}{\gamma_C}, \quad h_\mu = \frac{\mu_{e|r}}{\gamma_V}, \quad \gamma_C = \sqrt{1 + \hat{x} \cdot C_r \hat{x}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \quad (2)$$

We call h_w the *agreement field* — positive when the receiver already agrees with the message, large when it holds that view confidently — and h_μ the *distrust field*. In these variables the evidence the receiver assigns to the message is

$$Z = \Phi(h_w) + \Phi(h_\mu) - 2\Phi(h_w)\Phi(h_\mu), \quad (3)$$

with Φ the standard normal CDF. Z is small in the two *dissonant* quadrants — agreeing with a distrusted source, disagreeing with a trusted one — and large in the two consonant ones.

A Bayesian step on equation 1 leaves the exponential family; projecting the posterior back onto a Gaussian by maximum entropy, relative to the prior and subject to matching the first two moments, keeps the belief in the form equation 1 and yields a closed update in terms of derivatives of $\log Z$ (Oppen, 1996; 1998; Solla & Winther, 1999; Caticha, 2020). Appendix A carries the derivation. In the scaled fields,

$$\hat{w}_r \leftarrow \hat{w}_r + \frac{F_w}{\gamma_C} \sigma_e C_r \hat{x}, \quad C_r \leftarrow C_r + \frac{F_C}{\gamma_C^2} (C_r \hat{x})(C_r \hat{x})^\top, \quad (4)$$

$$\mu_{e|r} \leftarrow \mu_{e|r} + \frac{F_\mu}{\gamma_V} V_{e|r}, \quad V_{e|r} \leftarrow V_{e|r} + \frac{F_V}{\gamma_V^2} V_{e|r}^2, \quad (5)$$

where the four *modulation functions* are the log-derivatives of the evidence,

$$F_w = \frac{\partial \log Z}{\partial h_w} = (1 - 2\Phi(h_\mu)) \frac{\phi(h_w)}{Z}, \quad F_\mu = \frac{\partial \log Z}{\partial h_\mu} = (1 - 2\Phi(h_w)) \frac{\phi(h_\mu)}{Z}, \quad (6)$$

with $F_C = \partial^2 \log Z / \partial h_w^2 = -F_w(F_w + h_w)$ and $F_V = -F_\mu(F_\mu + h_\mu)$, and ϕ the standard normal density. Equations equation 1–equation 6 are inherited; everything from §4 onwards is new.

Three consequences, all of which the collective behaviour rests on. First, F_w carries the prefactor $1 - 2\Phi(h_\mu)$, which is *negative* for a distrusted emitter: the receiver moves its opinion *away* from what a distrusted source says. Effective antiferromagnetic couplings appear without being put in. Second, F_μ carries $1 - 2\Phi(h_w)$, so perceived agreement drives distrust down and perceived disagreement drives it up — agreement builds trust. This is the hinge that §4 turns. Third, F_C and F_V are negative almost everywhere, so both uncertainties shrink monotonically: **the dynamics anneals and never reaches a stationary state**. A society slows down rather than settling, so *when* a population is measured is a parameter of the experiment and not an incidental detail. Every quantity we report is at a fixed interaction count, stated in §5, and Appendix D reports the sensitivity of each to it.

What is and is not optimal. The update is optimal in a specific and limited sense, and it is worth being exact about which. It is an entropic (maximum-entropy Gaussian projection) approximation to a Bayesian online step, derived for a *single* noisy student–teacher channel in the thermodynamic limit (Oppen, 1996; Solla & Winther, 1999; Caticha, 2020); the identification of the inferred channel-noise level with distrust, and its promotion to a dynamical variable, is due to Alves & Caticha (2016). Deploying that rule in an N -agent population is out of distribution relative to its derivation: the receiver’s model of the world contains one teacher, and the population contains $N - 1$ mutually inconsistent ones. **We make no claim that the population dynamics is optimal in any sense.** What the paper studies is precisely the gap — the collective consequences of a locally well-motivated rule deployed where its derivation does not reach.

4 A CLASS-CORRELATED REPRESENTATION ERROR, AND WHAT IT DOES LOCALLY

Nothing in §3 refers to an agent’s class. We now let a fraction f_d of agents make one specific mistake, and show that its local consequence can be worked out exactly.

The error is in the representation of an interaction, not of an issue. A receiver scoring an incoming message forms a representation of the *interaction*: the issue, the opinion asserted, and — for a discriminating agent — one further coordinate that depends only on who sent it. Write $z_r \in \{0, 1\}$ for the discriminator indicator, drawn independently of class with $\mathbb{P}(z_r=1) = f_d$, and

$$\varphi_r(\hat{x}, \sigma_e, \kappa_e) = \left[\sigma_e \hat{x} / \gamma_C ; z_r s(\kappa_r, \kappa_e) \right] \in \mathbb{R}^{K+1}, \quad W_r = [\hat{w}_r ; d], \quad (7)$$

with $s(\kappa_r, \kappa_e) = \kappa_r \kappa_e$. The compatibility score $W_r \cdot \varphi_r$ is then exactly the agreement field of equation 2 shifted by a class-dependent constant,

$$H = h_w + D_{e|r}, \quad D_{e|r} = z_r d \kappa_r \kappa_e, \quad (8)$$

and the receiver evaluates equation 3–equation 6 at H in place of h_w . Two features of equation 7 are load-bearing and are easy to get wrong. Attaching the extra coordinate to the *issue* rather than the interaction would multiply the shift by σ_e and divide it by γ_C , so it would flip with the emitter’s stated opinion and decay as the agent grew confident — neither of which is intended. And the coordinate is deterministic, carrying no posterior variance, which is what leaves γ_C untouched; a learned coordinate of variance v_a would perturb the normalisation at order v_a/γ_C^2 . Appendix B gives the assumptions in full, together with the learned-prejudice variant.

The feature is task-irrelevant, and the population makes it relevant. The receiver models the label as $\sigma_T = \text{sign}(w_T \cdot \hat{x})$ for an unknown teacher w_T , so the class label is irrelevant in the exact sense $I(\sigma_T; \kappa_e | \hat{x}, w_T) = 0$: conditioned on the issue and the teacher, knowing the source’s class says nothing about the correct answer. In the population, weights and labels are drawn independently, so this holds at initialisation by construction. It does not continue to hold. Once opinions align with class, the label *does* predict what a source will say, and the spurious feature becomes self-fulfillingly valid — an agent audited against the realised data distribution would find no bias to remove. We return to this in §8.

Sign convention, stated once. $d > 0$ denotes **tolerance towards the in-group and intolerance towards the out-group**. This is forced by the algorithm rather than chosen: by equation 6 the factor $1 - 2\Phi(H)$ is negative when $H > 0$ and μ is *distrust*, so adding $+d$ makes a receiver read its counterpart as more agreeable than it is and grow more trusting. Adding $-d$ manufactures disagreement and breeds distrust. Earlier presentations of this model tabulate the opposite assignment while placing the discriminatory regime at $d > 0$ in their text and figures; the two are inconsistent, and taken literally they exchange the halves of the diagram. Appendix B works the discrepancy through and shows both readings.

4.1 THE LOCAL FLOW: AN EXACT SYMMETRY, EXACTLY BROKEN

Held at a fixed issue and emitter opinion, a receiver’s two fields move along (F_w, F_μ) evaluated at (H, h_μ) . Appendix A gives the exact increments, which carry additional terms in F_C and F_V ; those terms share the symmetry below and are cubic near the fixed point, so they change neither the symmetry nor the linearisation. We therefore state the result for the leading flow $X_D(h_w, h_\mu) = (F_w(H, h_\mu), F_\mu(H, h_\mu))$.

Proposition 1 (Rigid translation, exact symmetry, explicit separatrix). *Let Z, F_w, F_μ be as in equation 3–equation 6. Then*

- (i) **Translation.** $X_D(h_w, h_\mu) = X_0(h_w + D, h_\mu)$ for every D . The portrait at field D is the unbiased portrait translated rigidly by $-D$ along the agreement axis.
- (ii) **Nullclines.** F_w vanishes exactly on $\{h_\mu = 0\}$ and F_μ exactly on $\{H = 0\}$, so $(-D, 0)$ is the unique fixed point.
- (iii) **Symmetry at $D = 0$.** X_0 is equivariant under the point reflection $u \mapsto -u$ and under exchanging the two sectors.
- (iv) **Separatrix.** The line $h_\mu = h_w + D$ is globally invariant, and $(-D, 0)$ is a hyperbolic saddle with eigenvalues $\mp 2/\pi$. Below the line the flow runs to the consonant state agree and trust; above it, to disagree and distrust.

The proof is three lines of $\Phi(-a) = 1 - \Phi(a)$ and is given in Appendix A; every claim is also verified numerically in the accompanying code. Part (i) is stronger than a perturbative statement: there is no expansion in d anywhere. Part (iii) is the symmetry that the discrimination field breaks, and part (iv) says how — the basin of *agree and trust* is $\{h_\mu < h_w + D\}$, which grows strictly with D . A discriminating receiver therefore has an enlarged trust basin towards its own class and an enlarged distrust basin towards the other, and every such agent is biased towards the same alignment of distrust with class.

4.2 A PREDICTION WITH NO FREE PARAMETERS

Proposition 1(iv) says a channel is captured by the trust attractor exactly when its initial dissonance gap $\psi = h_\mu - h_w$ falls below D . Writing Ψ for the distribution of ψ at initialisation, the expected asymptotic trust on a channel carrying field D is $G(D) = 2\Psi(D) - 1$. Under equation 8 with balanced classes, a non-discriminating receiver contributes nothing on average because $G(0) = 0$, so the trust–class correlation of §5 obeys

$$C_{CT} = f_d G(d). \quad (9)$$

At the model’s own initialisation Ψ is available in closed form: $C = I$ and unit issues give $\gamma_C = \sqrt{2}$ and $h_w \sim \mathcal{N}(0, \frac{1}{2})$, while $V = 1$ and $\mu \sim \mathcal{U}(-1, 1)$ give h_μ uniform on $\pm 1/\sqrt{2}$, so ψ is the convolution of a uniform and a Gaussian of equal scale $1/\sqrt{2}$ and Ψ integrates elementarily (Appendix A). *Equation equation 9 therefore has no fitted quantity of any kind.* It predicts that C_{CT} is odd in d ; that dividing the measured map by f_d collapses it onto the single curve G ; that $|C_{CT}| \leq f_d$; and that the initial susceptibility is $\partial_d C_{CT}|_0 = f_d G'(0)$ with $G'(0) = 0.9655$.

Equation equation 9 is a *local* theory: it treats each channel as deciding on its own. §6 shows the measured correlation exceeds it, and we report the residual $\Delta = C_{CT} - f_d G(d)$ as the population’s own contribution — the amplification that comes from non-discriminating agents acquiring class-aligned trust second-hand.

4.3 WHY DISTRUST FOLLOWS CLASS BEFORE OPINION DOES

The field enters the two sectors asymmetrically, and the asymmetry is structural rather than quantitative. By equation 6, $\text{sign}(F_\mu) = -\text{sign}(H)$, so D can *flip the sign* of the trust update whenever $|h_w| < |D|$. But $\text{sign}(F_w)$ depends on h_μ alone: a receiver moves towards a trusted source’s opinion and away from a distrusted one’s whatever D is, and D only rescales the step. Class structure therefore reaches the opinion sector only *through* the trust sector. The direct representational drive on the opinion–class correlation is moreover of the opposite sign and roughly twenty-five times weaker — the relevant coefficients are $\mathbb{E}[F_C|w_e \cdot \hat{x}] = -0.0247$ against $\partial_H F_\mu|_0 = -2/\pi$ (Appendix A).

Corollary 1. $\text{sign}(C_{CT}) = \text{sign}(d)$, with $|C_{CT}| \geq |C_{CO}|$ throughout; C_{CO} is small and of the opposite sign at small $f_d|d|$, and becomes positive only once $f_d G(d)$ is large enough for the trust field to reorganise opinions.

The small- $|d|$ negative lobe in C_{CO} is a falsifiable prediction that no prior description of this model makes; §6 tests it.

There is no critical field strength for C_{CT} , and there cannot be. It is tempting to read the sharp sign change of C_{CT} at $d = 0$ as a transition and to look for a critical curve $d_c(f_d)$. That search cannot succeed. F_μ vanishes *identically* on the line $h_w = 0$, so $\partial_{h_\mu} F_\mu = 0$ there: the class-trust contrast has no linear restoring force at the class-symmetric state. The contrast is a marginally damped, driven response rather than an instability, and by the class-relabelling symmetry C_{CT} is an odd function of d with $C_{CT} \equiv 0$ at $d = 0$ for every N . The sign change is a *forced zero-crossing of an odd response*, not a bifurcation. This also explains, structurally, why an earlier semi-analytic closure of this model obtained a largest eigenvalue that was identically zero across its whole parameter grid: the vanishing is a property of the rule, not a shortcoming of that closure. Where a genuine threshold may live is the onset of *ideological* alignment, and §6 looks for it there.

5 SETUP, OBSERVABLES, AND HOW REGIMES ARE NAMED

Protocol. Unless stated otherwise: $N = 40$ agents in two equal classes, $K = 30$, agenda $P = 5$ (simple, $\alpha = P/K = 0.17$) or $P = 100$ (complex, $\alpha = 3.3$), the symmetric field equation 8, discriminators drawn independently of class. Initial state $\hat{w} \sim \mathcal{N}(0, I)$, $C = I$, $V = 1$ and $\mu \sim \mathcal{U}(-1, 1)$, the last chosen so that a fresh population starts with as many frustrated triples as unfrustrated ones. Emitter, receiver and issue are drawn uniformly at each step, and every society is run for 500 interactions per ordered pair. K is recovered rather than chosen and the interaction count is calibrated; Appendix D gives the provenance of every number, along with the sensitivity of each observable to the measurement time — which, since the dynamics anneals rather than converging (§3), is a genuine parameter and not a convergence tolerance.

Every reported quantity is a mean over 24 independent replicates per grid point, each with its own initial condition, agenda, class assignment, discriminator draw and interaction schedule, on a 49×49 grid in (d, f_d) . Uncertainties are replicate standard errors for per-point quantities. For anything derived from the map as a whole — a boundary location, a regime area, a crossover — we resample whole replicate sheets rather than propagating per-point errors, because replicates within a sheet share an interaction schedule and their errors are therefore correlated (Appendix D).

Observables. Two pairwise quantities carry everything. The *ideological overlap* $q_{IJ} = \cos(w_I, w_J)$ is $+1$ when two agents agree on every issue and -1 when they disagree on every one. The *directed trust* $t_{e|r} = 1 - 2\Phi(h_\mu)$ is $+1$ for a fully trusted source and -1 for a fully distrusted one. With the signed class relation $s_{IJ} = \kappa_I \kappa_J$ and $\bar{t}_{IJ} = (t_{I|J} + t_{J|I})/2$,

$$C_{CT} = \langle s_{IJ} \bar{t}_{IJ} \rangle, \quad C_{CO} = \langle s_{IJ} q_{IJ} \rangle, \quad C_{TO} = \langle \bar{t}_{IJ} q_{IJ} \rangle, \quad (10)$$

averaged over unordered pairs. These are a *renaming* of the three correlations used in earlier presentations of this model, not a correction: the quantities are numerically identical, and the accompanying code asserts equality to 10^{-12} . We rename them only because C_{CT} , C_{CO} , C_{TO} say which sectors are being correlated, and the earlier symbols do not.

The class correlations are not sufficient, and this matters for one whole region of the diagram.

A population can be sharply polarised along an axis unrelated to class, in which case all of equation 10 are near zero — and a class-only analysis would report it as structureless. We therefore also measure class-independent polarisation. Let λ_a be the eigenvalues of the centred overlap matrix and set

$$P = \lambda_1^2 / \sum_a \lambda_a^2, \quad \hat{P}_O = \frac{P - P_{\text{null}}}{P_{\text{max}}(N) - P_{\text{null}}}, \quad (11)$$

and identically $\hat{P}_T$ on the symmetrised trust matrix. P is largest for a two-bloc (rank-one) structure and about $1/\text{rank}$ for an isotropic one. The squared-eigenvalue fraction is used rather than a variance fraction because the centred trust matrix is indefinite, where a variance fraction is meaningless; using one formula for both sectors is what makes the comparison in §7 a comparison of like with like. Both the null P_{null} , measured at initialisation, and the attainable maximum $P_{\text{max}}(N) = (N-1)^2/((N-1)^2 + N-2)$ are N -dependent, so both are divided out; without that, the finite-size panels would show a drift in polarisation with N that is purely an artefact of the estimator.

We also measure social balance over triples (Heider, 1946; Cartwright & Harary, 1956), which separates *organised* disagreement from disorder: two coherent, mutually opposed blocs give balance 1 whatever they are made of, and a population that cannot settle goes negative. We report the continuous scores B_O , B_T and their sign-based counterparts separately, and only ever describe the latter as a fraction of balanced triples — a continuous product of overlaps is not one. Both are computed in closed form from matrix products rather than by enumerating triples (Appendix C).

Regimes are classified, not labelled by hand. Rather than placing names on a picture, we fix thresholds, apply them to each replicate separately, and report the modal regime together with the fraction of replicates that agree. Thresholds are calibrated from the no-discrimination null: the cut on $|C_{CT}|$ is three standard deviations of C_{CT} where $d = 0$, so “class-correlated” is measured against how large the correlation gets when there is provably no class signal, at the same N . Classifying replicates rather than the replicate mean matters — a point where half the replicates fall in one regime and half in another is bistable, and averaging first would render it as an intermediate value

and hide that. Points where fewer than 70% of replicates agree are drawn hatched, which is the boundary uncertainty made visible. Appendix C gives the thresholds, the rule, and the regime areas recomputed at $\{2/3, 1, 3/2\}$ times each cut.

On naming. Regime names describe measured behaviour. In particular we do not use “spin glass”: that name requires overlap distributions, metastability and initialisation dependence, none of which a pair correlation establishes, so the region with $d < 0$ is named for what is actually measured — negative affective balance with no ideological alignment. We likewise write “regimes” and “crossovers” rather than “phases” and “transitions” except where the finite-size evidence of §6 supports the stronger word.

6 COLLECTIVE REGIMES

[This section reports the replicated (d, f_d) map: the regime classification with boundary uncertainty, the three correlations with their standard errors, the test of the parameter-free prediction equation 9 and the amplification residual, the magnitude-matched controls, and the finite-size behaviour of the ideological-alignment onset. It awaits the main campaign, which is in progress.]

7 AGENDA COMPLEXITY AND THE ORDERING OF POLARISATION

[This section reports the first-passage analysis: the polarisation times of the two sectors against agenda complexity $\alpha = P/K$, the crossover with its confidence interval, and the sensitivity of both to the threshold. It awaits the corresponding runs.]

8 LIMITATIONS AND CONCLUSION

Limitations. The agents are single-layer perceptrons over a fixed shared embedding; the classes are binary, balanced and immutable; every pair may interact, so there is no network structure; the population is closed and of fixed size. Issues are isotropic synthetic vectors rather than embeddings of anything. Because the dynamics anneals rather than converging, every regime we report is a regime *at a stated interaction count*, and Appendix D shows how each observable moves with that count. “Trust” is a model construct — an inferred channel-noise level — and not a measurement of anything psychological. We have not validated any of this against behavioural or societal data, and it would not license conclusions about human groups if we had.

Conclusion. A task-irrelevant feature can do more than bias a single prediction. When learners also infer whom to trust, a class-correlated representation error changes the information flow of the entire population: it breaks a symmetry of the local dynamics, and the break is amplified into group-aligned collective structure that no individual agent chose and that the agenda’s complexity can reorder in time. Two consequences seem worth carrying away. The first is that the vulnerability comes from the very machinery that makes each agent a good inferential citizen — drawing conclusions about *sources* as well as about the world — so giving interacting agents a well-founded model of one another’s reliability is not a neutral improvement. The second is that the label starts out carrying no information and does not stay that way: once opinions align with class, class genuinely predicts opinion, and an agent audited against the realised distribution would find nothing to correct. Population-level measurement is not a refinement of per-agent evaluation here; it is the only thing that can see the effect at all.

REPRODUCIBILITY STATEMENT

Every figure and number in this paper is produced by the accompanying code, which contains the model, the observables, the sweep driver and the analysis, together with a test suite that checks the modulation functions against numerical derivatives of $\log Z$, the invariants of the dynamics, the observables against configurations whose values are known in closed form, and each analytical claim of §4 and Appendix A against its numerical evaluation. A regression test additionally pins the dynamics bit-for-bit against an independent earlier implementation.

Every random stream is derived by hashing an explicit run key, so a society’s trajectory depends only on that key and not on batch size, worker count or grid resolution; any single society in any figure can be re-run in isolation from the identifiers stored in the results file. Runs are checkpointed per chunk and resume by set difference. Per-replicate values are stored rather than averaged away, so every uncertainty in the paper can be recomputed, and the parameter table in Appendix D records the provenance of each value — recovered, calibrated, or chosen — along with the numerical safeguards the model does not itself prescribe.

ETHICS STATEMENT

This paper studies how discriminatory structure can arise among interacting learning agents, and two points bear stating.

The model is not a claim about people. Its agents are single-layer perceptrons, its groups are arbitrary inert labels, and its “tolerance” is a single scalar. We use the vocabulary of in-group and out-group because it names the structure compactly, not because the model licenses inferences about human prejudice, whose causes are historical, institutional and material in ways nothing here represents. No human subjects are involved and no personal data is used. The synthetic class labels are not proxies for any protected attribute, and results about them should not be read as results about any real demographic group.

The practical direction of the work is defensive. The result implies that a population of learning agents can occupy a discriminatory collective state while every agent in it would pass an individual audit, and that the state becomes invisible to precisely the evaluation protocol that is standard. That argues for population-level measurement of deployed multi-agent systems, and the order parameters used here need only a pairwise opinion alignment and a pairwise trust, both of which are cheap to elicit.

USE OF LARGE LANGUAGE MODELS

Large language models were used as a general-purpose assistive tool during this work: to help restructure and edit prose, to suggest organisation for the manuscript, and to assist with software engineering in the accompanying code. All modelling choices, derivations, experimental designs and claims are the authors’ own, and every analytical result stated in the paper is additionally verified numerically by tests released with the code. The authors take full responsibility for the contents of this paper, including any text or code that was drafted with such assistance.

REFERENCES

- Felippe Alves and Nestor Caticha. Sympatric multiculturalism in opinion models. *AIP Conference Proceedings*, 1757:060005, 2016.
- Tibor Antal, Pavel L. Krapivsky, and Sidney Redner. Dynamics of social balance on networks. *Physical Review E*, 72(3):036121, 2005.
- Robert Axelrod. The dissemination of culture: A model with local convergence and global polarization. *Journal of Conflict Resolution*, 41(2):203–226, 1997.
- Solon Barocas, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, 2023.
- Fabian Baumann, Philipp Lorenz-Spreen, Igor M. Sokolov, and Michele Starnini. Modeling echo chambers and polarization dynamics in social networks. *Physical Review Letters*, 124(4):048301, 2020.
- Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2021.
- Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. Language (technology) is power: A critical survey of “bias” in nlp. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

- Dorwin Cartwright and Frank Harary. Structural balance: a generalization of heider’s theory. *Psychological Review*, 63(5):277–293, 1956.
- Claudio Castellano, Santo Fortunato, and Vittorio Loreto. Statistical physics of social dynamics. *Reviews of Modern Physics*, 81(2):591–646, 2009.
- Nestor Caticha. Entropic dynamics in neural networks, the renormalization group and the hamilton–jacobi–bellman equation. *Entropy*, 22(5):587, 2020.
- Nestor Caticha and Felipe Alves. Trust, law and ideology in a nn agent model of the us appellate courts. In *European Symposium on Artificial Neural Networks (ESANN)*, 2019.
- Nestor Caticha and Renato Vicente. Agent-based social psychology: from neurocognitive processes to social data. *Advances in Complex Systems*, 14(5):711–731, 2011.
- Nestor Caticha, Jonatas Cesar, and Renato Vicente. For whom will the bayesian agents vote? *Frontiers in Physics*, 3:25, 2015.
- Guillaume Deffuant, David Neau, Frederic Amblard, and Gérard Weisbuch. Mixing beliefs among interacting agents. *Advances in Complex Systems*, 3(01n04):87–98, 2000.
- Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. In *Innovations in Theoretical Computer Science (ITCS)*, 2012.
- Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- Rainer Hegselmann and Ulrich Krause. Opinion dynamics and bounded confidence: models, analysis and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3), 2002.
- Fritz Heider. Attitudes and cognitive organization. *The Journal of Psychology*, 21(1):107–112, 1946.
- Seth A. Marvel, Jon Kleinberg, Robert D. Kleinberg, and Steven H. Strogatz. Continuous-time model of structural balance. *Proceedings of the National Academy of Sciences*, 108(5):1771–1776, 2011.
- Manfred Oppel. On-line versus off-line learning from random examples: General results. *Physical Review Letters*, 77(22):4671, 1996.
- Manfred Oppel. A bayesian approach to online learning. In David Saad (ed.), *On-line Learning in Neural Networks*. Cambridge University Press, 1998.
- Lucas S. Simões and Nestor Caticha. Mean field studies of a society of interacting agents. In *Bayesian Inference and Maximum Entropy Methods in Science and Engineering*, pp. 131–140. Springer, 2018.
- Sara A. Solla and Ole Winther. Optimal online learning: a bayesian approach. *Computer Physics Communications*, 121–122:94–97, 1999.

A THE LEARNING RULE, THE REDUCED FLOW, AND THE PROOFS

A.1 DERIVATION OF THE UPDATE

The receiver’s belief before an interaction is the product of Gaussians equation 1. It observes σ_e and treats it as a report of an unknown true label σ_T through a channel that flips with probability $\varepsilon(z)$. Marginalising over labels and over the issue as the emitter may have parsed it,

$$L(\sigma_e | \hat{x}, u) = \sum_{\sigma_T} \int P(\sigma_e | \sigma_T, y, \hat{x}, u) P(\sigma_T | y, \hat{x}, u) P(y | \hat{x}, u) dy, \quad (12)$$

with $P(\sigma_e | \sigma_T, z) = (1 - \varepsilon(z))\delta_{\sigma_e, \sigma_T} + \varepsilon(z)\delta_{\sigma_e, -\sigma_T}$ and $P(\sigma_T | y, w) = \Theta(\sigma_T y \cdot w)$. Two modelling choices keep the algebra Gaussian: taking $\varepsilon(z) = \Phi(z)$ with z Gaussian of mean $\mu_{e|r}$

and variance $V_{e|r}$ makes the flip probability a bounded function of a Gaussian variable while preserving conjugacy of the affective sector, and taking $P(y | \hat{x}, u)$ Gaussian with variance $\|w\|^{-2}$ lets an inexperienced agent be correspondingly unsure it has parsed the issue as the emitter did. Averaging L over the current belief collapses the Gaussian integrals onto the two scaled fields and gives equation 3: the evidence depends on the receiver's state only through (h_w, h_μ) . That is the origin of the sector symmetry in Proposition 1(iii).

The exact posterior is not Gaussian. Replacing it by the maximum-entropy Gaussian relative to the prior, subject to matching the first two moments, gives updates in terms of derivatives of $\log Z$, which in the scaled fields are equation 4–equation 5 with the modulation functions equation 6.

A.2 THE EXACT FIELD INCREMENTS, AND WHY THE REDUCED FLOW IS LEGITIMATE

The updates move $\hat{w}_r$ and $\mu_{e|r}$, while h_w and h_μ are scaled versions whose normalisations also move. Carrying the algebra through for a channel held at fixed $(\hat{x}, \sigma_e)$, and writing $a = \hat{x} \cdot C_r \hat{x} / \gamma_C^2$ and $b = V_{e|r} / \gamma_V^2$, both in $(0, 1)$:

$$\Delta h_w = a \left[F_w - \frac{a}{2} h_w F_C \right], \quad \Delta h_\mu = b \left[F_\mu - \frac{b}{2} h_\mu F_V \right], \quad (13)$$

with all modulation functions evaluated at (H, h_μ) . Three observations justify stating Proposition 1 for the leading flow.

The correction terms share the symmetry. F_C is even and h_w odd under $u \mapsto -u$, so $h_w F_C$ is odd, exactly like F_w . Nothing in Proposition 1(iii) depends on dropping them.

They are cubic at the fixed point. On the diagonal $F_w(s, s) = -(2/\pi)s + O(s^3)$ and $F_C(s, s) = O(s^2)$, so $h_w F_C = O(s^3)$. The linearisation, the eigenvalues and the tangent to the separatrix are unaffected.

The isotropic case is the model's own initial condition. With $C = I$, $V = 1$ and unit issues, $a = b = 1/2$ exactly. Anisotropy $a \neq b$ tilts the separatrix to slope $\sqrt{b/a}$ but preserves the hyperbolic saddle, so every conclusion survives by structural stability. Part (i) needs no assumption at all: D enters only through H , in equation 13 as much as in the leading flow.

A.3 PROOF OF PROPOSITION 1

(i) D appears in F_w and F_μ only through $H = h_w + D$, so $X_D(h_w, h_\mu) = X_0(h_w + D, h_\mu)$ by inspection.

(ii) F_w carries the factor $1 - 2\Phi(h_\mu)$, which vanishes exactly at $h_\mu = 0$, and ϕ/Z is finite and positive; F_μ carries $1 - 2\Phi(H)$, vanishing exactly at $H = 0$. The two nullclines meet only at $(-D, 0)$.

(iii) Take $D = 0$. From $\Phi(-z) = 1 - \Phi(z)$, $Z(-h_w, -h_\mu) = (1 - \Phi(h_w)) + (1 - \Phi(h_\mu)) - 2(1 - \Phi(h_w))(1 - \Phi(h_\mu)) = Z(h_w, h_\mu)$, while $1 - 2\Phi(-h_\mu) = -(1 - 2\Phi(h_\mu))$ and ϕ is even. Hence $F_w(-u) = -F_w(u)$ and likewise for F_μ , so $X_0(-u) = -X_0(u)$. Sector exchange is immediate from the symmetry of Z in its two arguments, which gives $F_w(x, y) = F_\mu(y, x)$.

(iv) On $h_\mu = H$ the two modulation functions coincide, since Z is symmetric and the prefactors exchange, so $F_w = F_\mu$ there and the flow is parallel to the line: it is invariant, not merely tangent. (On the antidiagonal $h_\mu = -H$ one gets $F_\mu = -F_w$, so that line is invariant too.) Linearising at $(-D, 0)$, the diagonal terms vanish by (ii) and $\partial_{h_\mu} F_w|_0 = \partial_H F_\mu|_0 = -2\phi(0)^2/Z(0, 0) = -2/\pi$, using $Z(0, 0) = 1/2$ and $\phi(0)^2 = 1/(2\pi)$. The Jacobian is $\begin{pmatrix} 0 & -2/\pi \\ -2/\pi & 0 \end{pmatrix}$, with eigenvalues $\mp 2/\pi$ and eigenvectors along the diagonal (stable) and antidiagonal (unstable). $\square$

A.4 THE GAP DISTRIBUTION AND G

At initialisation $\hat{w} \sim \mathcal{N}(0, I_K)$ and $\hat{x}$ is a unit vector, so $\hat{w}_r \cdot \hat{x} \sim \mathcal{N}(0, 1)$ and, with $C = I$ giving $\gamma_C = \sqrt{2}$, $h_w \sim \mathcal{N}(0, \frac{1}{2})$. With $V = 1$, $\gamma_V = \sqrt{2}$ and $\mu \sim \mathcal{U}(-1, 1)$, h_μ is uniform on $(-c, c)$ with $c = 1/\sqrt{2}$. The gap $\psi = h_\mu - h_w$ is therefore the convolution of $\mathcal{U}(-c, c)$ with $\mathcal{N}(0, \sigma^2)$,

$\sigma = 1/\sqrt{2}$. Using the antiderivative $\mathcal{F}(z) = z\Phi(z) + \phi(z)$ of Φ ,

$$\Psi(t) = \frac{\sigma}{2c} \left[\mathcal{F}\left(\frac{t+c}{\sigma}\right) - \mathcal{F}\left(\frac{t-c}{\sigma}\right) \right], \quad G(D) = 2\Psi(D) - 1. \quad (14)$$

G is odd, strictly increasing and tends to ± 1 ; its slope at the origin is $G'(0) = 2p_\psi(0) = [\Phi(c/\sigma) - \Phi(-c/\sigma)]/c = 0.9655$. The accompanying code checks Ψ against 2×10^6 Monte Carlo samples to within 2×10^{-3} .

A.5 THE CHANNEL ASYMMETRY OF COROLLARY 1

$\text{sign}(F_\mu) = \text{sign}(1 - 2\Phi(H)) = -\text{sign}(H)$, so the field can invert the trust update whenever $|h_w| < |D|$ and the signs oppose. By contrast $\text{sign}(F_w) = \text{sign}(1 - 2\Phi(h_\mu))$ does not involve D at all, so the direction of every opinion update is set by trust alone. To first order in d , expanding $F_w(h_w + D, \cdot) = F_w + D F_C + O(D^2)$ and averaging over the initial measure at a class-symmetric state with $C_r = cI$, the class-correlated drift of the opinion overlap is proportional to $\mathbb{E}[F_C | w_e \cdot \hat{x}]$, which evaluates to -0.0247 : negative, and some twenty-five times smaller in magnitude than the trust-channel coefficient $-2/\pi$. Hence C_{CO} is driven *negative* at small $f_d|d|$ and turns positive only when the trust field is strong enough to reorganise opinions — a level set of $f_d G(d)$, which is approximately hyperbolic at small d and flattens as G saturates.

A.6 WHY NO CRITICAL FIELD STRENGTH EXISTS FOR C_{CT}

$F_\mu = (1 - 2\Phi(H))\phi(h_\mu)/Z$ carries a factor that vanishes identically on $H = 0$. Hence $F_\mu \equiv 0$ along that entire line and so, in particular, $\partial_{h_\mu} F_\mu|_{H=0} = 0$: linearising the affective sector about the class-symmetric state gives a Jacobian with a zero eigenvalue in the class-contrast direction. The contrast obeys $\frac{d}{d\tau}(\mu_{\text{out}} - \mu_{\text{in}}) = J_\mu(\mu_{\text{out}} - \mu_{\text{in}}) + b(f_d, d)$ with $J_\mu = 0$, i.e. a driven response with no restoring force, not an eigenvalue crossing. Together with the class-relabelling symmetry — which forces C_{CT} to be odd in d and exactly zero at $d = 0$ for every N — this establishes that the sign change at $d = 0$ cannot be a bifurcation. A finite-size collapse performed around it would be a collapse about a smooth odd function and would manufacture an exponent with no referent.

B THE AUGMENTED REPRESENTATION, AND THE SIGN CONVENTION

B.1 ASSUMPTIONS UNDER WHICH EQUATION 7 REDUCES EXACTLY

Write the augmented weights as $W_r = [\hat{w}_r; a_r]$ with a Gaussian belief of variance v_a on the extra coordinate. The general compatibility score is

$$H^{\text{aug}} = \frac{h_w + \sigma_e a_r z_r s / \gamma_C}{\sqrt{1 + z_r v_a / \gamma_C^2}} = h_w + \frac{D}{\gamma_C} - \frac{h_w z_r v_a}{2\gamma_C^2} + O(v_a^2), \quad (15)$$

so equation 8 is recovered exactly under four assumptions, which we state rather than leave implicit.

A1 (deterministic coordinate, $v_a = 0$). The class coordinate is part of the representation, not a learned parameter, so it carries no posterior variance. This is what leaves γ_C untouched and suppresses any covariance update along that direction. It is the only assumption needed for exactness; relaxing it gives the $O(v_a/\gamma_C^2)$ correction above.

A2 (fixed prejudice). The weight on the coordinate is the hyperparameter d and is not updated. Relaxing A2 gives a *learned-prejudice* variant in which $a_r \leftarrow a_r + (F_w/\gamma)v_a z_r s$, so the strength of the bias would itself adapt to how well it predicted. That is a natural extension and we do not study it.

A3 (interaction-level, not issue-level). The coordinate attaches to the message, because κ_e is a property of the source and not of the issue. This is what keeps the shift from being multiplied by σ_e and divided by γ_C — either of which would make the bias flip with the emitter’s stated opinion, or decay as the receiver grew confident.

A4 (bounded binary feature). $s \in \{0, \pm 1\}$, so the shift is bounded by $|d|$ and there are exactly six inequivalent fillings of the 2×2 matrix, tabulated below.

B.2 THE SIX DISCRIMINATION PATTERNS

Rows index the receiver’s class, columns the emitter’s; $+d$ is the tolerant entry and $-d$ the intolerant one.

case	D/d	who discriminates	how
1	$\begin{pmatrix} +1 & 0 \\ 0 & 0 \end{pmatrix}$	A only	in-group favouritism
2	$\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$	A only	out-group hostility
3	$\begin{pmatrix} +1 & -1 \\ 0 & 0 \end{pmatrix}$	A only	both
4	$\begin{pmatrix} 0 & 0 \\ +1 & 0 \end{pmatrix}$	both	in-group favouritism
5	$\begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix}$	both	out-group hostility
6	$\begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}$	both	both

Case 6, used throughout the main text, is the one in which the two classes play symmetric roles, so any asymmetry appearing in the results is spontaneous rather than imposed. Compactly, $D_{e|r} = z_r d \kappa_r \kappa_e$.

B.3 THE SIGN CONVENTION, AND AN INCONSISTENCY IN EARLIER PRESENTATIONS

Three statements are in play in earlier accounts of this model: that a discriminating receiver evaluates its update at $h_w + D_{e|r}$; that for $d > 0$ the in-group entries of the matrix are $-d$; and that the discriminatory regime lies at $d > 0$. The first two together imply the opposite of the third.

The argument is one line of equation 6. F_μ carries the factor $1 - 2\Phi(H)$, which is negative whenever $H > 0$, and μ is *distrust*, updated as $\mu \leftarrow \mu + F_\mu V / \gamma_V$. So perceived agreement drives μ down and builds trust, and perceived disagreement drives it up. A receiver adding $+d$ therefore reads its counterpart as more agreeable than it is and grows *more trusting*: $+d$ is the tolerant entry. Assigning $-d$ to the in-group means a society at $d > 0$ distrusts its own class, which places the discriminatory regime at $d < 0$ and mirrors every map in d .

Any one of three repairs resolves it, and they are the same change: flip the table so in-group entries are $+d$ (what we do, since it leaves the field equation untouched and preserves the meaning of every published axis); or keep the table and write the field equation with a minus; or keep both and relabel $d < 0$ as the discriminatory direction. The choice has no effect on the physics — the same regimes exist in the same places — only on which half of the axis carries which name. The accompanying code implements both readings behind a flag and asserts that they are exact mirror images.

C OBSERVABLES AND THE CLASSIFICATION RULE

Conventions. We use the signed class relation $s_{IJ} = \kappa_I \kappa_J \in \{-1, +1\}$. A $\{0, 1\}$ indicator, which counts only same-class pairs, cannot represent anti-correlation: under perfect reverse alignment it returns $-(N/2 - 1)/(N - 1)$ rather than -1 , and its range depends on N . Since half the diagram has the field favouring the out-group, the signed indicator is the appropriate one. Directed trust has $t_{II} \equiv +1$, which the closed form below assumes.

Closed form for the balance aggregates. Enumerating triples costs $O(N^3)$ per society, which is prohibitive across a replicated grid. For any M with unit diagonal,

$$\sum_{\substack{(I,J,K) \\ \text{distinct, ordered}}} M_{IJ} M_{JK} M_{KI} = \text{tr}(M^3) - 3\left(\sum_{I,J} M_{IJ} M_{JI} - N\right) - N, \quad (16)$$

and each unordered triple appears six times, three as the forward cycle and three as the reverse. For symmetric M this gives the ideological balance; for the asymmetric trust matrix the two orientations are exactly the two terms of the affective balance, so one expression serves both. The identity is verified against enumeration in the test suite, for continuous and sign matrices alike.

Finite-size values of the sanity checks. Two configurations are worth recording because the natural expectation is wrong at finite N . If every pair trusts every other and all opinions align, then

with balanced classes $C_{CT} = C_{CO} = -1/(N - 1)$, not 0 — averaging s_{IJ} over unordered pairs of a balanced split leaves a deficit of one pair’s worth. And the polarisation equation 11 cannot reach 1: zeroing the diagonal of a rank-one ss^T leaves a spectrum $N - 1$ once and -1 with multiplicity $N - 2$ after centring, so $P_{\max}(N) = (N - 1)^2 / ((N - 1)^2 + N - 2)$, which is 0.94 at $N = 16$ and 0.995 at $N = 200$. Both values are asserted in the tests, and $P_{\max}$ is divided out so that polarisation is comparable across N .

The classification rule. With thresholds θ_c on $|C_{CT}|$, θ_p on the standardised polarisations and θ_b on B_T , each replicate is assigned in this fixed order: no class correlation and no polarisation in either sector $\rightarrow$ *weakly structured*; no class correlation but polarisation present $\rightarrow$ *polarized, class-uncorrelated*; class correlation of the wrong sign $\rightarrow$ *counter-aligned, frustrated*; class correlation with opinion following class $\rightarrow$ *discriminatory, ideological*; otherwise $\rightarrow$ *discriminatory, class-dominant*. Thresholds are three null standard deviations of the corresponding statistic measured at $d = 0$ at the same N , so they carry the finite-size noise floor and need no re-tuning by eye. The modal label over replicates is reported together with the agreement fraction, and regime areas are recomputed at $\{2/3, 1, 3/2\}$ times each threshold.

We considered clustering the order-parameter vectors instead and rejected it: cluster identity is not stable across N , across agenda size, or between a baseline and its controls, and comparing exactly those things is the purpose of the finite-size and control figures.

D SIMULATION DETAILS

[Parameter table with provenance, pseudocode, seeding scheme, compute resources, the measurement-time sensitivity of every observable, and the replicate-resampling procedure used for map-level uncertainties. Awaits the campaign.]